

API TESTING REPORT

Functional & Validation Testing — Cardiac Patient Monitoring API

Postman/Swagger-Based Testing of Auth, Patients, Vital Signs, Medications & Appointments

1. Introduction

This report documents the functional and validation testing performed on the CardiacPatientMonitoringApi, an ASP.NET Core Web API prototype for monitoring cardiac patients. Testing covered the authentication and authorization layer (Register, Login, JWT, Current User, Authorization) as well as full CRUD and validation behavior for the Patients, Vital Signs, Medications, and Appointments modules.

All requests were executed manually using Postman/Swagger against the running API, with each test case designed to verify a specific expected HTTP status code and behavior — including success paths, validation failures, missing/invalid data, and not-found scenarios.

2. Objective

- Confirm that each endpoint returns the correct HTTP status code for both valid and invalid requests.
- Confirm that registration, login, JWT authentication, and role-based authorization behave as expected.
- Verify complete CRUD behavior (Create, Read, Update, Delete) for Patients, Vital Signs, Medications, and Appointments.
- Verify that invalid or out-of-range data (e.g. an implausible heart rate, a missing medication start date) is rejected rather than silently accepted.
- Verify that requests referencing non-existent resources (patients, vital signs, medications, appointments) correctly return 404 Not Found.
- Document evidence of each test case, including a dedicated space for the Postman/Swagger screenshot of the request and response.

3. Testing Environment

Project	CardiacPatientMonitoring.Api
Technology	ASP.NET Core Web API, C#/.NET, Entity Framework Core, SQL Server, ASP.NET Core Identity, JWT Authentication
Testing Tool	Postman / Swagger (manual requests, one request per test case)
Modules Under Test	Auth (Register/Login/JWT/Current User), Authorization, Patients, Vital Signs, Medications, Appointments
Total Test Cases	65
Date	Aug/23/2026

4. Test Cases — Summary

The table below lists all test cases executed against the API. Each case verifies a specific expected outcome for a given endpoint, covering both success and failure paths.

#	Module	Test Case	Endpoint	Expected
01	Register	Register with valid data	POST /api/Auth/Register	201 Created
02	Register	Register with missing username/email	POST /api/Auth/Register	400 Bad Request
03	Register	Register with missing password	POST /api/Auth/Register	400 Bad Request
04	Register	Register with weak/invalid password	POST /api/Auth/Register	400 Bad Request
05	Register	Register with an already existing user	POST /api/Auth/Register	400 Bad Request
06	Login	Login with valid credentials	POST /api/Auth/Login	200 OK + JWT token
07	Login	Login with wrong password	POST /api/Auth/Login	401 Unauthorized
08	Login	Login with non-existing user	POST /api/Auth/Login	401 Unauthorized
09	Login	Login with missing credentials	POST /api/Auth/Login	400 Bad Request
10	JWT	Access protected endpoint without token	GET /api/[protected resource]	401 Unauthorized
11	JWT	Access protected endpoint with invalid token	GET /api/[protected resource]	401 Unauthorized
12	JWT	Access protected endpoint with valid token	GET /api/[protected resource]	200 OK
13	Current User	Get current authenticated user	GET /api/Auth/Me	200 OK + user information
14	Current User	Get current user without JWT	GET /api/Auth/Me	401 Unauthorized
15	Authorization	User accesses authorized endpoint	GET /api/[role-restricted resource]	200 OK
16	Authorization	User accesses endpoint without required permission/role	GET /api/[role-restricted resource]	403 Forbidden
17	Patients	Get all patients	GET /api/Patients	200 OK

#	Module	Test Case	Endpoint	Expected
18	Patients	Get existing patient	GET /api/Patients/{id}	200 OK
19	Patients	Get patient that doesn't exist	GET /api/Patients/{id}	404 Not Found
20	Patients	Create valid patient	POST /api/Patients	201 Created
21	Patients	Create patient with missing required field	POST /api/Patients	400 Bad Request
22	Patients	Update existing patient	PUT /api/Patients/{id}	200 OK
23	Patients	Update non-existing patient	PUT /api/Patients/{id}	404 Not Found
24	Patients	Delete existing patient	DELETE /api/Patients/{id}	204 No Content
25	Patients	Delete non-existing patient	DELETE /api/Patients/{id}	404 Not Found
26	Vital Signs	Get all vital signs	GET /api/VitalSigns	200 OK
27	Vital Signs	Get existing vital sign	GET /api/VitalSigns/{id}	200 OK
28	Vital Signs	Get non-existing vital sign	GET /api/VitalSigns/{id}	404 Not Found
29	Vital Signs	Get vital signs for existing patient	GET /api/VitalSigns/patient/{patientId}	200 OK
30	Vital Signs	Get vital signs for non-existing patient	GET /api/VitalSigns/patient/{patientId}	404 Not Found
31	Vital Signs	Create valid vital sign	POST /api/VitalSigns	201 Created
32	Vital Signs	Create vital sign for invalid patient	POST /api/VitalSigns	400 Bad Request
33	Vital Signs	Create vital sign with invalid heart rate (500)	POST /api/VitalSigns	400 Bad Request
34	Vital Signs	Update existing vital sign	PUT /api/VitalSigns/{id}	200 OK
35	Vital Signs	Update non-existing vital sign	PUT /api/VitalSigns/{id}	404 Not Found
36	Vital Signs	Delete existing vital sign	DELETE /api/VitalSigns/{id}	204 No Content
37	Vital Signs	Delete non-existing vital sign	DELETE /api/VitalSigns/{id}	404 Not Found
38	Medications	Get medications for existing patient	GET /api/Medications/patient/{patientId}	200 OK

#	Module	Test Case	Endpoint	Expected
39	Medications	Get medications for non-existing patient	GET /api/Medications/patient/{patientId}	404 Not Found
40	Medications	Search medications	GET /api/Medications/search	200 OK
41	Medications	Get existing medication by ID	GET /api/Medications/{id}	200 OK
42	Medications	Get non-existing medication	GET /api/Medications/{id}	404 Not Found
43	Medications	Create valid medication	POST /api/Medications	201 Created
44	Medications	Create medication for invalid patient	POST /api/Medications	404 Not Found
45	Medications	Create medication without name	POST /api/Medications	400 Bad Request
46	Medications	Create medication without StartDate	POST /api/Medications	400 Bad Request
47	Medications	Update existing medication	PUT /api/Medications/{id}	200 OK
48	Medications	Update non-existing medication	PUT /api/Medications/{id}	404 Not Found
49	Medications	Delete existing medication	DELETE /api/Medications/{id}	204 No Content
50	Medications	Delete non-existing medication	DELETE /api/Medications/{id}	404 Not Found
51	Appointments	Get all appointments	GET /api/Appointments	200 OK
52	Appointments	Filter appointments by PatientId	GET /api/Appointments?patientId={id}	200 OK
53	Appointments	Filter appointments by Status	GET /api/Appointments?status={status}	200 OK
54	Appointments	Filter appointments by DoctorName	GET /api/Appointments?doctorName={name}	200 OK
55	Appointments	Get appointments for existing patient	GET /api/Appointments/patient/{patientId}	200 OK
56	Appointments	Get appointments for non-existing patient	GET /api/Appointments/patient/{patientId}	404 Not Found
57	Appointments	Get existing appointment	GET /api/Appointments/{id}	200 OK
58	Appointments	Get non-existing appointment	GET /api/Appointments/{id}	404 Not Found
59	Appointments	Create valid appointment	POST /api/Appointments	201 Created

#	Module	Test Case	Endpoint	Expected
60	Appointments	Create appointment for invalid patient	POST /api/Appointments	404 Not Found
61	Appointments	Create appointment with missing required field	POST /api/Appointments	400 Bad Request
62	Appointments	Update existing appointment	PUT /api/Appointments/{id}	200 OK
63	Appointments	Update non-existing appointment	PUT /api/Appointments/{id}	404 Not Found
64	Appointments	Delete existing appointment	DELETE /api/Appointments/{id}	204 No Content
65	Appointments	Delete non-existing appointment	DELETE /api/Appointments/{id}	404 Not Found

Endpoint	POST /api/Auth/Register
Module	Register
Rule / Behavior Tested	Username/email is required
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Register / Register - Missing Full Name

POST `{{baseUrl}}/api/Auth/register` **Send**

Docs Params Authorization Headers (9) Body **Scripts** Settings Cookies

Pre-request

Post-response

```

1 pm.test("Status code is 400", function () {
2   pm.response.to.have.status(400);
3 });
4
5 pm.test("FullName validation error is returned", function () {
6   const jsonData = pm.response.json();
7
8   pm.expect(jsonData).to.have.property("errors");
9   pm.expect(jsonData.errors).to.have.property("FullName");
10 });

```

400 Bad Request • 7 ms • 415 B • Save Response

{ } JSON Preview Debug with AI

```

1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3   "title": "One or more validation errors occurred.",
4   "status": 400,
5   "errors": {
6     "FullName": [
7       "The FullName field is required."
8     ]
9   },
10  "traceId": "00-a436addce655728c47919be030b6b3f1-d9cac7ffe3155e0e-00"
11 }

```

TC-03 Register with missing password

Endpoint	POST /api/Auth/Register
Module	Register
Rule / Behavior Tested	Password is required
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 01 - Authentic... / Register / **Register - Missing Required Fields** Save Share

POST ▼ `{{baseUrl}}/api/Auth/register` Send ▼

Docs Params Authorization Headers (9) **Body** ● Scripts ● Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ▼ Schema Beautify

```

1  {
2
3  }

```

Body Cookies Headers (4) Test Results (2/2) ↺ **400 Bad Request** • 6 ms • 634 B • 🌐 Save Response ⋮

{} **JSON** ▼ ▶ Preview 🔍 Debug with AI ▼ 🔄 🔍 📄 🔗

```

1  {
2      "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3      "title": "One or more validation errors occurred.",
4      "status": 400,
5      "errors": {
6          "Email": [
7              "The Email field is required.",
8              "The Email field is not a valid e-mail address."
9          ],
10         "FullName": [
11             "The FullName field is required."
12         ],
13         "Password": [
14             "The Password field is required.",
15             "The field Password must be a string or array type with a minimum length of '6'."
16         ]
17     },
18     "traceId": "00-723c1ec18df5745dc337e8e3b28d1d68-b36e83bf33d5612a-00"
19 }

```

TC-04 Register with weak/invalid password

Endpoint	POST /api/Auth/Register
Module	Register
Rule / Behavior Tested	Password must meet complexity requirements
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Register / Register - Short Password

POST `{{baseUrl}}/api/Auth/register` **Send**

Docs Params Authorization Headers (9) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Schema Beautify

```

1 {
2   "email": "shortpassword@cardiacapi.com",
3   "password": "Ab1",
4   "fullName": "Short Password Test"
5 }

```

Body Cookies Headers (4) Test Results (2/2) **400 Bad Request** 5 ms 463 B Save Response

{ } JSON Preview Debug with AI

```

1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3   "title": "One or more validation errors occurred.",
4   "status": 400,
5   "errors": {
6     "Password": [
7       "The field Password must be a string or array type with a minimum length of '6'."
8     ]
9   },
10  "traceId": "00-86258f29e6304b8f7a36e56d25f26fe1-40ee9bae3c9b3d63-00"
11 }

```

TC-05 Register with an already existing user

Endpoint	POST /api/Auth/Register
Module	Register
Rule / Behavior Tested	Username/email must be unique
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 01 - Authentication / Register / Register - Duplicate Email Save Share

POST ▼ `{{baseUrl}}/api/Auth/register` Send ▼

Docs Params Authorization Headers (9) Body ● Scripts ● Settings Cookies

Pre-request

Post-response ●

```

1 pm.test("Status code is 409", function () {
2   pm.response.to.have.status(409);
3 });
4
5 pm.test("Duplicate email message is returned", function () {
6   const jsonData = pm.response.json();
7
8   pm.expect(jsonData.message).to.eql(
9     "An account with this email already exists."
10  );
11 });

```

Packages </> Snippets  

Body Cookies Headers (4) Test Results (2/2)  **409 Conflict** • 94 ms • 210 B •   Save Response ...

{ } JSON ▼ ▷ Preview  Debug with AI ▼     

```

1 {
2   "message": "An account with this email already exists."
3 }

```

Login — Test Cases

TC-06 Login with valid credentials

Endpoint	POST /api/Auth/Login
Module	Login
Rule / Behavior Tested	Valid credentials authenticate successfully
Expected Result	200 OK + JWT token
Actual Result	200 OK + JWT token — PASS

Evidence / Screenshot

Share 

▼ `{{baseUrl}} /api/Auth/login`

Cookies

Beautify



000

① ②

Endpoint	POST /api/Auth/Login
Module	Login
Rule / Behavior Tested	Password must match
Expected Result	401 Unauthorized
Actual Result	401 Unauthorized — PASS

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HTTP Cardiac Patient Monitoring / 01 - Authentication / Login / Login - Wrong Password Save Share

POST `{{baseUrl}}/api/Auth/login` Send

Docs Params Authorization Headers (9) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Schema Beautify

```

1 {
2   "email": "doctor1@cardiacapi.com",
3   "password": "WrongPassword@123"
4 }

```

Body Cookies Headers (4) Test Results (2/2) 401 Unauthorized • 138 ms • 198 B • Save Response

{} JSON Preview Pass the correct auth credentials

```

1 {
2   "message": "Invalid email or password."
3 }

```

TC-08 Login with non-existing user

Endpoint	POST /api/Auth/Login
Module	Login
Rule / Behavior Tested	User must exist
Expected Result	401 Unauthorized
Actual Result	401 — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 01 - Authentication / Login / Login - Unknown Email Save Share

POST ▼ `{{baseUrl}}/api/Auth/login` Send ▼

Docs Params Authorization Headers (9) **Body** ● Scripts ● Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ▼ Schema Beautify

```

1 {
2   "email": "unknown@cardiacapi.com",
3   "password": "Admin@12345"
4 }

```

Body Cookies Headers (4) Test Results (2/2) 🔄 **401 Unauthorized** • 8 ms • 198 B • 🌐 | Save Response ⋮

{} **JSON** ▼ ▶ Preview ⚙️ Pass the correct auth credentials ▼ 🔍 📄 🔗

```

1 {
2   "message": "Invalid email or password."
3 }

```

TC-09 Login with missing credentials

Endpoint	POST /api/Auth/Login
Module	Login
Rule / Behavior Tested	Username/email and password are required
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 01 - Authentication / Login / **Login - Invalid Email** Save Share

POST `{{baseUrl}}/api/Auth/login` Send

Docs Params Authorization Headers (9) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Schema Beautify

```

1  {
2
3  }

```

Body Cookies Headers (4) Test Results (0/2) 400 Bad Request • 40 ms • 505 B • Save Response

JSON Preview Debug with AI

```

1  {
2    "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3    "title": "One or more validation errors occurred.",
4    "status": 400,
5    "errors": {
6      "Email": [
7        "The Email field is required.",
8        "The Email field is not a valid e-mail address."
9      ],
10     "Password": [
11       "The Password field is required."
12     ]
13   },
14   "traceId": "00-0056d3a0b1cfb9ddc91e2d4d7d37ab-d2ab956087073689-00"
15 }

```

JWT — Test Cases

TC-10 Access protected endpoint without token

Endpoint	GET /api/[protected resource]
Module	JWT
Rule / Behavior Tested	Authentication token is required
Expected Result	401 Unauthorized
Actual Result	401 Unauthorized — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Invalid Token

GET {{baseUrl}} /api/Auth/me

Send

Docs Params **Authorization** Headers (7) Body Scripts Settings Cookies

Auth Type
No Auth

No Auth

This request does not use any authorization. ⓘ

Body Cookies Headers (4) Test Results 401 Unauthorized • 12 ms • 128 B • Save Response

Raw Preview Pass the correct auth credentials

1

TC-11 Access protected endpoint with invalid token

Endpoint	GET /api/[protected resource]
Module	JWT
Rule / Behavior Tested	Token must be valid/well-formed
Expected Result	401 Unauthorized
Actual Result	401 Unauthorized — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Invalid Token Save Share

GET `{{baseUrl}}/api/Auth/me` Send

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type

Bearer Token Token 👁️ ⚠️

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results 🕒 **401 Unauthorized** • 102 ms • 150 B • 🌐 Save Response ⋮

Raw Preview ⚙️ Pass the correct auth credentials 🔍 📄 🔗

1

TC-12 Access protected endpoint with valid token

Endpoint	GET /api/[protected resource]
Module	JWT
Rule / Behavior Tested	Valid token grants access
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Valid Token

Save Share

GET `{{baseUrl}}/api/Auth/me` Send

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type: Be... Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results 200 OK • 151 ms • 276 B Save Response

{ } JSON Preview Visualize

```

1  {
2    "id": "524e9544-df56-499e-9395-fcf69704213f",
3    "email": "admin@cardiacapi.com",
4    "fullName": "System Administrator",
5    "roles": [
6      "ADMIN"
7    ]
8  }

```

Current User — Test Cases

TC-13 Get current authenticated user

Endpoint	GET /api/Auth/Me
Module	Current User
Rule / Behavior Tested	Valid JWT returns current user info
Expected Result	200 OK + user information
Actual Result	200 OK + user information — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Valid Token Save Share

GET ▼ `{{baseUrl}}/api/Auth/me` Send ▼

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type
Be... ▼

Token
..... 👁 🔒

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results ↺ 200 OK • 151 ms • 276 B • 🌐 Save Response ⋮

{} JSON ▼ ▶ Preview 📐 Visualize ▼ 🔍 📄 🔗

```

1  {
2    "id": "524e9544-df56-499e-9395-fcf69704213f",
3    "email": "admin@cardiacapi.com",
4    "fullName": "System Administrator",
5    "roles": [
6      "ADMIN"
7    ]
8  }

```

TC-14 Get current user without JWT

Endpoint	GET /api/Auth/Me
Module	Current User
Rule / Behavior Tested	Authentication token is required
Expected Result	401 Unauthorized
Actual Result	401 Unauthorized — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Invalid Token

GET `{{baseUrl}}/api/Auth/me` **Send**

Docs Params **Authorization** Headers (7) Body Scripts Settings Cookies

Auth Type
No Auth

No Auth

This request does not use any authorization. ⓘ

Body Cookies Headers (4) Test Results **401 Unauthorized** • 12 ms • 128 B • Save Response

Raw Preview Pass the correct auth credentials

1

Authorization — Test Cases

TC-15 User accesses authorized endpoint

Endpoint	GET /api/[role-restricted resource]
Module	Authorization
Rule / Behavior Tested	User with correct role/permission is granted access
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Valid Token Save Share

GET ▼ `{{baseUrl}}/api/Auth/me` Send ▼

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type
Be... ▼

Token
..... 👁 🔒

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results ↺ 200 OK • 151 ms • 276 B • 🌐 Save Response ⋮

{} JSON ▼ ▶ Preview 📄 Visualize ▼ 🔍 📄 🔗

```

1  {
2    "id": "524e9544-df56-499e-9395-fcf69704213f",
3    "email": "admin@cardiacapi.com",
4    "fullName": "System Administrator",
5    "roles": [
6      "ADMIN"
7    ]
8  }

```

TC-16 User accesses endpoint without required permission/role

Endpoint	GET /api/[role-restricted resource]
Module	Authorization
Rule / Behavior Tested	User without correct role/permission is denied
Expected Result	403 Forbidden
Actual Result	403 Forbidden — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 01 - Authentication / Current User / Me - Invalid Token Save Share

GET `{{baseUrl}}/api/Auth/me` Send

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type
 Bearer Token Token 👁️ ⚠️

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results 🕒 **401 Unauthorized** • 102 ms • 150 B • 🌐 Save Response ⋮

Raw Preview ⚙️ Pass the correct auth credentials 🔍 📄 🔗

1

Patients — Test Cases

TC-17 Get all patients

Endpoint	GET /api/Patients
Module	Patients
Rule / Behavior Tested	Returns list of all patients
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / **Get All Patients** Save Share

GET `{{baseUrl}}/api/Patients` Send

Auth Type
Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token

Body **Cookies** **Headers (4)** **Test Results** 200 OK 100 ms 1.13 KB Save Response

JSON Preview Visualize

```

1  [
2    {
3      "id": 1,
4      "firstName": "Ahmad",
5      "lastName": "Hassan",
6      "dateOfBirth": "1985-04-12T00:00:00",
7      "gender": "Male",
8      "phoneNumber": "0599000001",
9      "createdAt": "2026-08-16T20:40:06.9884862"
10   },
11   {
12     "id": 5,
13     "firstName": "Maya",
14     "lastName": "Khaled",
15     "dateOfBirth": "1995-06-20T00:00:00",
16     "gender": "Female",
17     "phoneNumber": "0599000030",
18     "createdAt": "2026-08-16T21:30:21.9428741"
19   },
20   {
21     "id": 2

```

TC-18 Get existing patient

Endpoint	GET /api/Patients/{id}
Module	Patients
Rule / Behavior Tested	Returns patient by valid ID
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / Get Patient - Valid ID

GET `{{baseUrl}}/api/Patients/1` **Send**

Auth Type: Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token:

Body Cookies Headers (4) Test Results **200 OK** • 26 ms • 317 B Save Response

{ } JSON Preview Visualize

```

1 {
2   "id": 1,
3   "firstName": "Ahmad",
4   "lastName": "Hassan",
5   "dateOfBirth": "1985-04-12T00:00:00",
6   "gender": "Male",
7   "phoneNumber": "0599000001",
8   "createdAt": "2026-08-16T20:40:06.9884862"
9 }

```

TC-19 Get patient that doesn't exist

Endpoint	GET /api/Patients/{id}
Module	Patients
Rule / Behavior Tested	Patient ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / **Get Patient - Not Found** Save Share

GET `{{baseUrl}}/api/Patients/99` Send

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type
Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token

Body Cookies Headers (4) Test Results 404 Not Found • 45 ms • 202 B Save Response

`{}` JSON Preview Debug with AI

```

1 {
2   "message": "Patient with ID 99 was not found."
3 }
```

TC-20 Create valid patient

Endpoint	POST /api/Patients
Module	Patients
Rule / Behavior Tested	Valid patient data is accepted
Expected Result	201 Created
Actual Result	201 Created — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / Create Patient - Max Length

POST `{{baseUrl}}/api/Patients` **Send**

Docs Params Authorization Headers (10) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON [Schema](#) [Beautify](#)

```

1 {
2   "firstName": "AAAAAAAAAAAAAAAAAAAAAAAAAAAA",
3   "lastName": "Patient",
4   "dateOfBirth": "1995-05-15",
5   "gender": "Male",
6   "phoneNumber": "0599112233"
7 }

```

Body Cookies Headers (5) Test Results **201 Created** • 220 ms • 397 B • [Save Response](#)

`{}` JSON [Preview](#) [Visualize](#)

```

1 {
2   "id": 9,
3   "firstName": "AAAAAAAAAAAAAAAAAAAAAAAAAAAA",
4   "lastName": "Patient",
5   "dateOfBirth": "1995-05-15T00:00:00",
6   "gender": "Male",
7   "phoneNumber": "0599112233",
8   "createdAt": "2026-08-23T07:12:25.3388701Z"
9 }

```

TC-21 Create patient with missing required field

Endpoint	POST /api/Patients
Module	Patients
Rule / Behavior Tested	Required fields must be present
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / Create Patient - Missing Last Name

POST `{{baseUrl}}/api/Patients` **Send**

Auth Type: Bearer Token

Token:

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results **400 Bad Request** • 9 ms • 534 B • Save Response

`{}` JSON Preview Debug with AI

```

1  {
2    "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3    "title": "One or more validation errors occurred.",
4    "status": 400,
5    "errors": {
6      "FirstName": [
7        "The FirstName field is required.",
8        "The field FirstName must be a string or array type with a minimum length of '2'.",
9        "'First Name' must not be empty."
10     ]
11   },
12   "traceId": "00-de33f5a7689e01d0038c09bc45fe0fbc-bf756dfd64ea749a-00"
13 }

```

TC-22 Update existing patient

Endpoint	PUT /api/Patients/{id}
Module	Patients
Rule / Behavior Tested	Valid update data is accepted
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / Update Patient - Valid

PUT `{{baseUrl}}/api/Patients/6` **Send**

Auth Type: Bearer Token

Token:

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results **200 OK** • 102 ms • 324 B • Save Response

{ } JSON Preview Visualize

```

1 {
2   "id": 6,
3   "firstName": "Ahmad Updated",
4   "lastName": "Hassan",
5   "dateOfBirth": "1985-04-12T00:00:00",
6   "gender": "Male",
7   "phoneNumber": "0599000001",
8   "createdAt": "2026-08-17T11:06:16.165701"
9 }

```

TC-23 Update non-existing patient

Endpoint	PUT /api/Patients/{id}
Module	Patients
Rule / Behavior Tested	Patient ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 02 - Patients / Update Patient - Not Found Save Share

PUT ▼ `{{baseUrl}}/api/Patients/9999` Send ▼

Docs Params Authorization Headers (10) Body Scripts Settings Cookies

Auth Type

Bearer Token ▼

Token 👁 🔒

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results 🔄 404 Not Found 9 ms 204 B 🌐 📄 Save Response ⋮

{} JSON ▼ ▶ Preview 🐞 Debug with AI ▼ 🔍 📄 🔗

```
1 {
2   "message": "Patient with ID 9999 was not found."
3 }
```

TC-24 Delete existing patient

Endpoint	DELETE /api/Patients/{id}
Module	Patients
Rule / Behavior Tested	Valid ID is deleted
Expected Result	204 No Content
Actual Result	204 No Content — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 02 - Patients / Delete Patient - Valid

Save Share

DELETE `{{baseUrl}}/api/Patients/7` Send

Docs Params **Authorization** Headers (8) Body Scripts Settings Cookies

Auth Type
Bearer Token

Token
.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (2) Test Results 204 No Content 44 ms 81 B Save Response

Raw Preview Visualize

1

TC-25 Delete non-existing patient

Endpoint	DELETE /api/Patients/{id}
Module	Patients
Rule / Behavior Tested	Patient ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

HTTP

Cardiac Patient Monitoring / 02 - Patients / Delete Patient - Not Found

Save

Share

DELETE

{{baseUrl}}/api/Patients/88

Send

Docs

Params

Authorization

Headers (8)

Body

Scripts

Settings

Cookies

Auth Type

Bearer Token

Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body

Cookies

Headers (4)

Test Results

404 Not Found

9 ms

202 B

Save Response

{ } JSON

Preview

Debug with AI

1 {

2 | "message": "Patient with ID 88 was not found."

3 }

Vital Signs — Test Cases

TC-26 Get all vital signs	
Endpoint	GET /api/VitalSigns
Module	Vital Signs
Rule / Behavior Tested	Returns list of all vital signs
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / **Get All Vital Signs** Save Share

GET `{{baseUrl}}/api/VitalSigns` Send

Auth Type
Be... Token

The authorization header will be automatically generated when you send the request. Learn more about [earer Token](#) authorization.

Body Cookies Headers (4) Test Results 200 OK 75 ms 2.03 KB Save Response

{ } JSON Preview Visualize

```

1  [
2    {
3      "id": 6,
4      "patientId": 1,
5      "heartRate": 75,
6      "systolicPressure": 120,
7      "diastolicPressure": 80,
8      "oxygenSaturation": 98.50,
9      "recordedAt": "2026-08-21T15:00:00",
10     "notes": "Routine monitoring"
11   },
12   {
13     "id": 7,
14     "patientId": 1,
15     "heartRate": 75,
16     "systolicPressure": 250,
17     "diastolicPressure": 80,
18     "oxygenSaturation": 98.50,
19     "recordedAt": "2026-08-21T15:00:00",
20     "notes": "Systolic pressure too high"
21   }
22 ]

```

TC-27 Get existing vital sign

Endpoint	GET /api/VitalSigns/{id}
Module	Vital Signs
Rule / Behavior Tested	Returns vital sign by valid ID
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / Get Vital Sign - Valid ID

GET `{{baseUrl}}/api/VitalSigns/2` **Send**

Auth Type: Bearer Token

Token:

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results **200 OK** • 28 ms • 328 B • Save Response

{ } JSON Preview Visualize

```

1  {
2    "id": 2,
3    "patientId": 2,
4    "heartRate": 82,
5    "systolicPressure": 125,
6    "diastolicPressure": 82,
7    "oxygenSaturation": 97.50,
8    "recordedAt": "2026-08-16T19:40:07.2674371",
9    "notes": "Routine monitoring"
10 }
```

TC-28 Get non-existing vital sign

Endpoint	GET /api/VitalSigns/{id}
Module	Vital Signs
Rule / Behavior Tested	Vital sign ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / Get Vital Sign - Not Found

Save

Share

GET

{{baseUrl}}/api/VitalSigns/999

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

Cookies

Auth Type

Bearer Token

Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

BodyCookiesHeaders (4)Test Results

404 Not Found • 10 ms • 206 B • Save Response

{ } JSON

Preview

Debug with AI

1 {

2 "message": "Vital sign with ID 999 was not found."

3 }

TC-29 Get vital signs for existing patient

Endpoint	GET /api/VitalSigns/patient/{patientId}
Module	Vital Signs
Rule / Behavior Tested	Returns vital signs for a valid patient
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / **Get Patient Vital Signs** Save Share

GET `{{baseUrl}}/api/VitalSigns/patient/3` Send

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Auth Type
Be... ▼

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token 👁 🔒

Body Cookies Headers (4) Test Results ↺ 200 OK 42 ms 329 B 🌐 📄 Save Response ⋮

{} JSON ▼ ▶ Preview 🖼 Visualize ▼ 🔍 📄 🔗

```

1  [
2    {
3      "id": 3,
4      "patientId": 3,
5      "heartRate": 88,
6      "systolicPressure": 135,
7      "diastolicPressure": 85,
8      "oxygenSaturation": 96.00,
9      "recordedAt": "2026-08-16T20:10:07.2674384",
10     "notes": "Follow-up reading"
11   }
12 ]

```

TC-30 Get vital signs for non-existing patient

Endpoint	GET /api/VitalSigns/patient/{patientId}
Module	Vital Signs
Rule / Behavior Tested	Patient ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

HTTP

Cardiac Patient Monitoring / 03 - Vital Signs / Get Patient Vital Signs - Patient Not Found

Save

Share

GET

{{baseUrl}}/api/VitalSigns/patient/333

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

Auth Type

Bearer Token

Token

.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body

Cookies

Headers (4)

Test Results

404 Not Found

7 ms

203 B

Save Response

{}

JSON

Preview

Debug with AI

1 {

2 | "message": "Patient with ID 333 was not found."

3 }

TC-31 Create valid vital sign

Endpoint	POST /api/VitalSigns
Module	Vital Signs
Rule / Behavior Tested	Valid vital sign data is accepted
Expected Result	201 Created
Actual Result	201 Created — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / Create Vital Sign - Valid

POST {{baseUrl}}/api/VitalSigns

Auth Type: Bearer Token

Token:

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (5) Test Results

201 Created • 190 ms • 377 B • Save Response

```

1  {
2    "id": 12,
3    "patientId": 1,
4    "heartRate": 75,
5    "systolicPressure": 120,
6    "diastolicPressure": 80,
7    "oxygenSaturation": 98.5,
8    "recordedAt": "2026-08-21T15:00:00Z",
9    "notes": "Routine monitoring"
10 }
```

TC-32 Create vital sign for invalid patient

Endpoint	POST /api/VitalSigns
Module	Vital Signs
Rule / Behavior Tested	Referenced patient must exist
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / Invalid Patient ID

Save

Share

POST

{{baseUrl}} /api/VitalSigns

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

Auth Type

Bearer Token

Token

.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body

Cookies

Headers (4)

Test Results

400 Bad Request

11 ms

208 B

Save Response

{ } JSON

Preview

Debug with AI

1 {

2 "message": "Patient with ID 99999 does not exist."

3 }

TC-33 Create vital sign with invalid heart rate (500)

Endpoint	POST /api/VitalSigns
Module	Vital Signs
Rule / Behavior Tested	Heart rate must fall within a physiologically valid range
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS <i>Initially returned 200 OK (bug) — validation rule added, retested, now returns 400 Bad Request.</i>

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / Heart Rate Too High

SaveShare

POST{{baseUrl}}/api/VitalSigns

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

Cookies

Auth Type

Bearer Token

Token

he authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

BodyCookiesHeaders (4)Test Results

400 Bad Request10 ms477 BSave Response

JSONPreviewDebug with AI

```
1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3   "title": "One or more validation errors occurred.",
4   "status": 400,
5   "errors": {
6     "HeartRate": [
7       "The field HeartRate must be between 30 and 220.",
8       "Heart rate must be between 30 and 220 BPM."
9     ]
10  },
11   "traceId": "00-608121defb9768e5e7b0c3115500b395-cfb94723ba4e3440-00"
12 }
```

TC-34 Update existing vital sign

Endpoint	PUT /api/VitalSigns/{id}
Module	Vital Signs
Rule / Behavior Tested	Valid update data is accepted
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 03 - Vital Signs / Update Vital Sign - Valid

PUT `{{baseUrl}}/api/VitalSigns/10` **Send**

Docs Params Authorization Headers (10) **Body** Scripts Settings Cookies

none form-data x-www-form-urlencoded **raw** binary GraphQL JSON Schema Beautify

```

1 {
2   "patientId": 1,
3   "heartRate": 80,
4   "systolicPressure": 125,
5   "diastolicPressure": 82,
6   "oxygenSaturation": 97.5,
7   "recordedAt": "2026-08-21T14:00:00Z",
8   "notes": "Updated vital sign"
9 }

```

Body Cookies Headers (4) Test Results **200 OK** • 24 ms • 321 B • Save Response

{ } JSON Preview Visualize

```

1 {
2   "id": 10,
3   "patientId": 1,
4   "heartRate": 80,
5   "systolicPressure": 125,
6   "diastolicPressure": 82,
7   "oxygenSaturation": 97.5,
8   "recordedAt": "2026-08-21T14:00:00Z",
9   "notes": "Updated vital sign"
10 }

```

TC-35 Update non-existing vital sign

Endpoint	PUT /api/VitalSigns/{id}
Module	Vital Signs
Rule / Behavior Tested	Vital sign ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

 Cardiac Patient Monitoring / 03 - Vital Signs / Update Vital Sign - Valid

Save

Share

PUT

{{baseUrl}}/api/VitalSigns/1000

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

```
1 {
2   "patientId": 1,
3   "heartRate": 80,
4   "systolicPressure": 125,
5   "diastolicPressure": 82,
6   "oxygenSaturation": 97.5,
7   "recordedAt": "2026-08-21T14:00:00Z",
8   "notes": "Updated vital sign"
9 }
```

Body

Cookies

Headers (4)

Test Results

404 Not Found • 12 ms • 207 B • Save Response

{ } JSON

Preview

Debug with AI

```
1 {
2   "message": "Vital sign with ID 1000 was not found."
3 }
```

TC-36 Delete existing vital sign

Endpoint	DELETE /api/VitalSigns/{id}
Module	Vital Signs
Rule / Behavior Tested	Valid ID is deleted
Expected Result	204 No Content
Actual Result	204 No Content — PASS

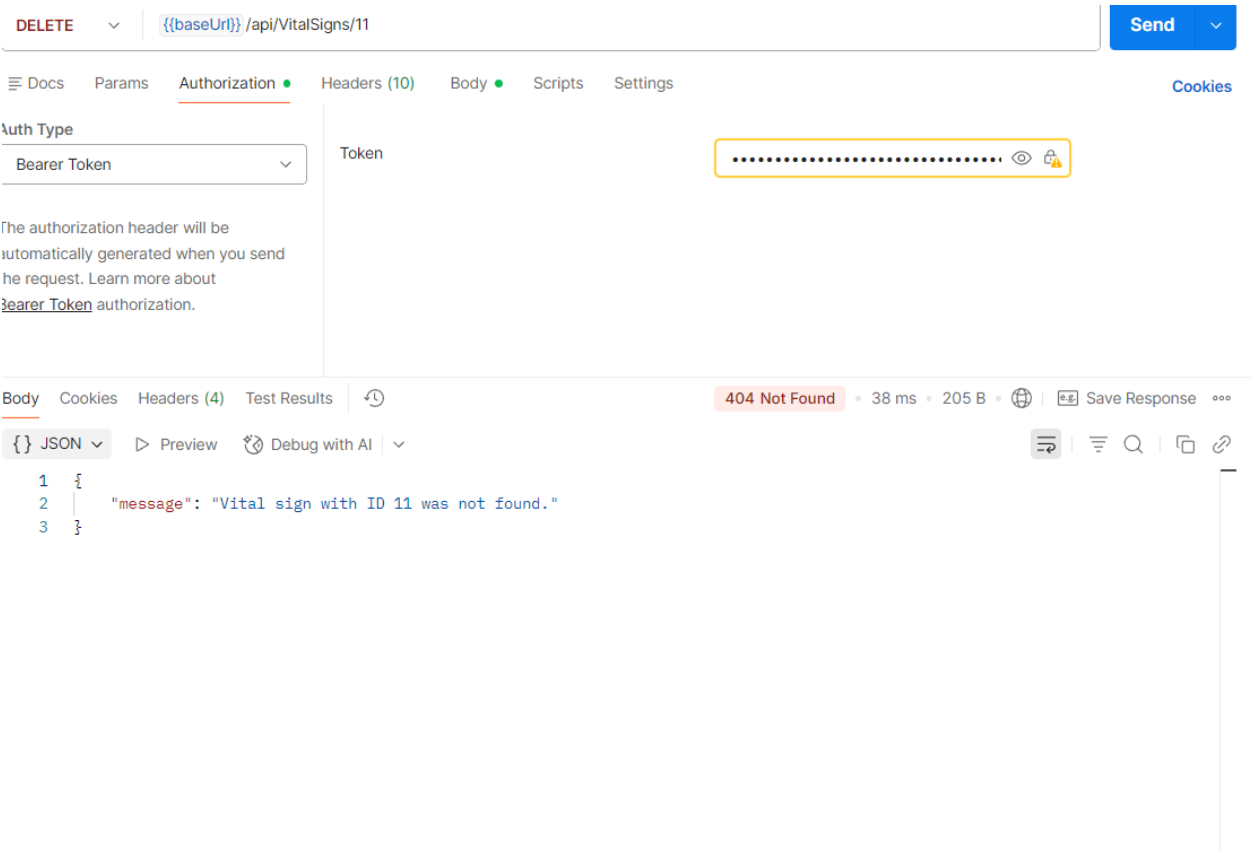
Evidence / Screenshot

The screenshot shows a REST client interface for a Cardiac Patient Monitoring API. The breadcrumb path is "Cardiac Patient Monitoring / 03 - Vital Signs / Delete Vital Sign - Valid". The request method is "DELETE" and the URL is "{{baseUri}}/api/VitalSigns/10". The "Authorization" tab is selected, showing "Auth Type" as "Bearer Token" and a "Token" field with a masked value. A note states: "The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization." The response status is "204 No Content" with a response time of 45 ms and a size of 81 B. The response body is empty.

TC-37 Delete non-existing vital sign

Endpoint	DELETE /api/VitalSigns/{id}
Module	Vital Signs
Rule / Behavior Tested	Vital sign ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot



Medications — Test Cases

TC-38 Get medications for existing patient	
Endpoint	GET /api/Medications/patient/{patientId}
Module	Medications
Rule / Behavior Tested	Returns medications for a valid patient
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Get Patient Medications

GET `{{baseUrl}}/api/Medications/patient/1` Send

Auth Type: Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token:

Body: 200 OK • 106 ms • 1.54 KB • Save Response

```

1 [
2   {
3     "id": 5,
4     "patientId": 1,
5     "name": "Lisinopril",
6     "dosage": "10 mg",
7     "frequency": "Once daily",
8     "startDate": "2026-08-21T00:00:00",
9     "endDate": null,
10    "notes": "Take in the morning"
11  },
12  {
13    "id": 6,
14    "patientId": 1,
15    "name": "Lisinopril",
16    "dosage": "20 mg",
17    "frequency": "Once daily",
18    "startDate": "2026-08-21T00:00:00",
19    "endDate": null,
20    "notes": "This is an intentionally very long medication note used to verify that the API correctly rejects notes
      that exceed the maximum allowed length for a medication record "
  }
]

```

TC-39 Get medications for non-existing patient

Endpoint	GET /api/Medications/patient/{patientId}
Module	Medications
Rule / Behavior Tested	Patient ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Get Patient Medications - Patient Not Found

GET

{{baseUrl}}/api/Medications/patient/9999

Send

DocsParamsAuthorizationHeaders (8)BodyScriptsSettings

Auth Type

Bearer Token

Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

BodyCookiesHeaders (4)Test Results

404 Not Found10 ms204 B

Save Response

JSON

Preview

Debug with AI

1{

2 "message": "Patient with ID 9999 was not found."

3 }

TC-40 Search medications

Endpoint	GET /api/Medications/search
Module	Medications
Rule / Behavior Tested	Returns medications matching search term
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Get Medication

GET `{{baseUrl}}/api/Medications/1` **Send**

Auth Type: Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token:

200 OK • 22 ms • 321 B

```

1  {
2    "id": 1,
3    "patientId": 1,
4    "name": "Aspirin",
5    "dosage": "100 mg",
6    "frequency": "Once daily",
7    "startDate": "2026-08-01T00:00:00",
8    "endDate": null,
9    "notes": "Updated medication instructions"
10 }
```

TC-41 Get existing medication by ID

Endpoint	GET /api/Medications/{id}
Module	Medications
Rule / Behavior Tested	Returns medication by valid ID
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Get Medication

GET `{{baseUrl}}/api/Medications/1` **Send**

Auth Type: Bearer Token

Token:

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body: 200 OK • 22 ms • 321 B

```

1  {
2    "id": 1,
3    "patientId": 1,
4    "name": "Aspirin",
5    "dosage": "100 mg",
6    "frequency": "Once daily",
7    "startDate": "2026-08-01T00:00:00",
8    "endDate": null,
9    "notes": "Updated medication instructions"
10 }
```

TC-42 Get non-existing medication

Endpoint	GET /api/Medications/{id}
Module	Medications
Rule / Behavior Tested	Medication ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Get Medication

SaveShare

GET{{baseUrl}}/api/Medications/100000

Send

DocsParamsAuthorizationHeaders (8)BodyScriptsSettings

Cookies

Auth Type

Bearer Token

Token

.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

BodyCookiesHeaders (4)Test Results

404 Not Found11 ms209 BSave Response

{ } JSON

PreviewDebug with AI

1 {2 | "message": "Medication with ID 100000 was not found."3 }

TC-43 Create valid medication

Endpoint	POST /api/Medications
Module	Medications
Rule / Behavior Tested	Valid medication data is accepted
Expected Result	201 Created
Actual Result	201 Created — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Create Medication - Valid

Save Share

POST {{baseUrl}}/api/Medications Send

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Auth Type
Bearer Token

Token
.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (5) Test Results 201 Created 93 ms 370 B Save Response

{ } JSON Preview Visualize

```

1 {
2   "id": 10,
3   "patientId": 1,
4   "name": "Lisinopril",
5   "dosage": "10 mg",
6   "frequency": "Once daily",
7   "startDate": "2026-08-21T00:00:00Z",
8   "endDate": null,
9   "notes": "Take in the morning"
10 }
```

TC-44 Create medication for invalid patient

Endpoint	POST /api/Medications
Module	Medications
Rule / Behavior Tested	Referenced patient must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Invalid Patient ID

SaveShare

POST{{baseUrl}}/api/MedicationsSend

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

Auth Type
Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token
.....

BodyCookiesHeaders (4)Test Results

404 Not Found • 21 ms • 205 B • Save Response

{ } JSONPreviewDebug with AI

```
1 {
2   "message": "Patient with ID 99999 was not found."
3 }
```

TC-45 Create medication without name

Endpoint	POST /api/Medications
Module	Medications
Rule / Behavior Tested	Name is required
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Missing Name

POST{{baseUrl}}/api/Medications

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

noneform-datax-www-form-urlencoderawbinaryGraphQLJSON

SchemaBeautify

```
1 {
2   "patientId": 1,
3   "dosage": "10 mg",
4   "frequency": "Once daily",
5   "startDate": "2026-08-21T00:00:00Z",
6   "notes": "Missing medication name"
7 }
```

BodyCookiesHeaders (4)Test Results

400 Bad Request6 ms438 B

Save Response

JSONPreviewDebug with AI

```
1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3   "title": "One or more validation errors occurred.",
4   "status": 400,
5   "errors": {
6     "Name": [
7       "The Name field is required.",
8       "Medication name is required."
9     ]
10  },
11   "traceId": "00-5b8e2555e617987402893420492c6edc-2f8fc30fc6d919cf-00"
12 }
```

TC-46 Create medication without StartDate

Endpoint	POST /api/Medications
Module	Medications
Rule / Behavior Tested	StartDate is required
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS <i>Initially returned 200 OK and defaulted StartDate to 0001-01-01T00:00:00 (bug) — DTO validation fixed, retested, now returns 400 Bad Request.</i>

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Missing StartDate

POST `{{baseUrl}}/api/Medications` **Send**

Docs Params Authorization Headers (10) **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** Schema Beautify

```

1 {
2   "patientId": 1,
3   "name": "Lisinopril",
4   "dosage": "10 mg",
5   "frequency": "Once daily",
6   "notes": "Missing start date"
7 }

```

Body Cookies Headers (4) Test Results **400 Bad Request** 4 ms 417 B Save Response

JSON Preview Debug with AI

```

1 {
2   "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3   "title": "One or more validation errors occurred.",
4   "status": 400,
5   "errors": {
6     "StartDate": [
7       "The StartDate field is required."
8     ]
9   },
10  "traceId": "00-febc15c164efe2822edaab0511186f2c-cb9d9d093214334c-00"
11 }

```

TC-47 Update existing medication

Endpoint	PUT /api/Medications/{id}
Module	Medications
Rule / Behavior Tested	Valid update data is accepted
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

HTTP Cardiac Patient Monitoring / 04 - Medications / **Update Medication - Valid** Save Share

PUT `{{baseUrl}}/api/Medications/6` Send

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Auth Type
Bearer Token Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) Test Results 200 OK 70 ms 459 B Save Response

`{}` JSON Preview Visualize

```

1  {
2    "id": 6,
3    "patientId": 1,
4    "name": "Lisinopril",
5    "dosage": "10 mg",
6    "frequency": "Once daily",
7    "startDate": "2026-08-21T00:00:00Z",
8    "endDate": null,
9    "notes": "This is an intentionally very long medication note used to verify that the API correctly rejects notes that
10   exceed the maximum allowed length for a medication record."
  }

```

TC-48 Update non-existing medication

Endpoint	PUT /api/Medications/{id}
Module	Medications
Rule / Behavior Tested	Medication ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

[HTTP](#) Cardiac Patient Monitoring / 04 - Medications / Update Medication - Not Found

PUT

{{baseUrl}}/api/Medications/9999

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

```
1 {
2   "patientId": 1,
3   "name": "Lisinopril",
4   "dosage": "10 mg",
5   "frequency": "Once daily",
6   "startDate": "2026-08-21T00:00:00Z",
7   "notes": "This is an intentionally very long medication note used to verify that the API correctly rejects notes that
8             exceed the maximum allowed length for a medication record."
9 }
```

Body

Cookies

Headers (4)

Test Results

404 Not Found • 8 ms • 207 B • Save Response

JSON

Preview

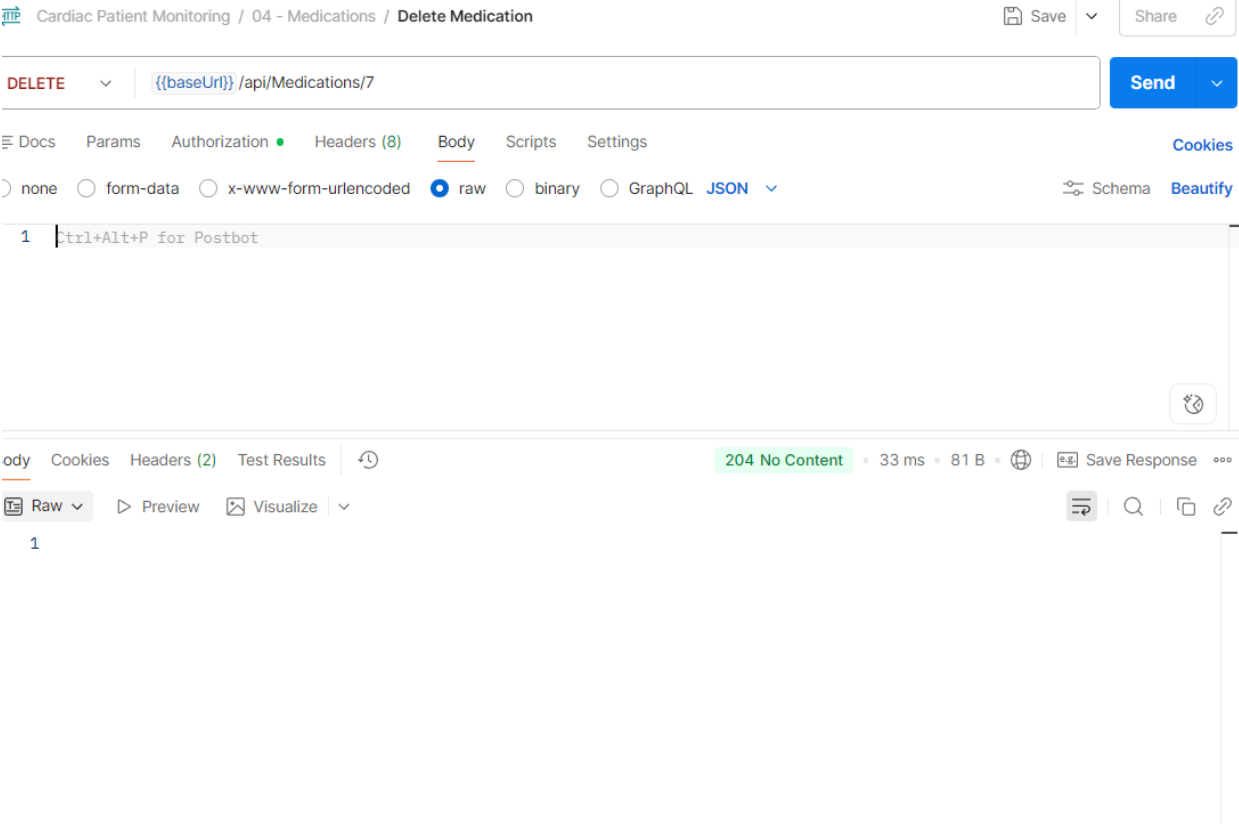
Debug with AI

```
1 {
2   "message": "Medication with ID 9999 was not found."
3 }
```

TC-49 Delete existing medication

Endpoint	DELETE /api/Medications/{id}
Module	Medications
Rule / Behavior Tested	Valid ID is deleted
Expected Result	204 No Content
Actual Result	204 No Content — PASS

Evidence / Screenshot



TC-50 Delete non-existing medication

Endpoint	DELETE /api/medications/{id}
Module	Medications
Rule / Behavior Tested	Medication ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 04 - Medications / Delete Medication

SaveShare

DELETE{{baseUrl}}/api/Medications/8

Send

DocsParamsAuthorizationHeaders (10)BodyScriptsSettings

Auth TypeBe...Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

BodyCookiesHeaders (4)Test Results

404 Not Found23 ms204 BSave Response

{ } JSONPreviewDebug with AI

```
1 {
2   "message": "Medication with ID 8 was not found."
3 }
```

Appointments — Test Cases

TC-51 Get all appointments	
Endpoint	GET /api/Appointments
Module	Appointments
Rule / Behavior Tested	Returns list of all appointments
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / **Get All Appointments** Save Share

GET ▼ `{{baseUrl}}/api/Appointments` Send ▼

Docs Params Authorization Headers (10) Body Scripts Settings Cookies

Auth Type
Be... ▼

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token 👁 🔒

Body Cookies Headers (4) Test Results 🕒 200 OK • 121 ms • 634 B 🌐 📄 Save Response ⋮

{} JSON ▼ ▶ Preview 🖼 Visualize ▼ 🔍 📄 🔗

```

1  [
2    {
3      "id": 2,
4      "patientId": 2,
5      "appointmentDate": "2026-09-20T10:00:00",
6      "doctorName": "Dr. Ahmad Saleh",
7      "reason": "Updated cardiac follow-up",
8      "status": "Scheduled",
9      "notes": null
10   },
11   {
12     "id": 7,
13     "patientId": 1,
14     "appointmentDate": "2026-09-15T10:00:00",
15     "doctorName": "Dr. Ahmad Saleh",
16     "reason": "Cardiac follow-up",
17     "status": "Scheduled",
18     "notes": null
19   },
20   {
21     "id": 4

```

TC-52 Filter appointments by PatientId

Endpoint	GET /api/Appointments?patientId={id}
Module	Appointments
Rule / Behavior Tested	Filters correctly by PatientId
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / **Get Appointments by Patient** Save Share

ET ▼ `{{baseUrl}}/api/Appointments/patient/1` Send ▼

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Type
Bearer Token ▼

Token 👁 🔒

authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

200 OK • 43 ms • 469 B • 🌐 Save Response ⋮

JSON ▼ ▶ Preview 🖼 Visualize ⌵

```

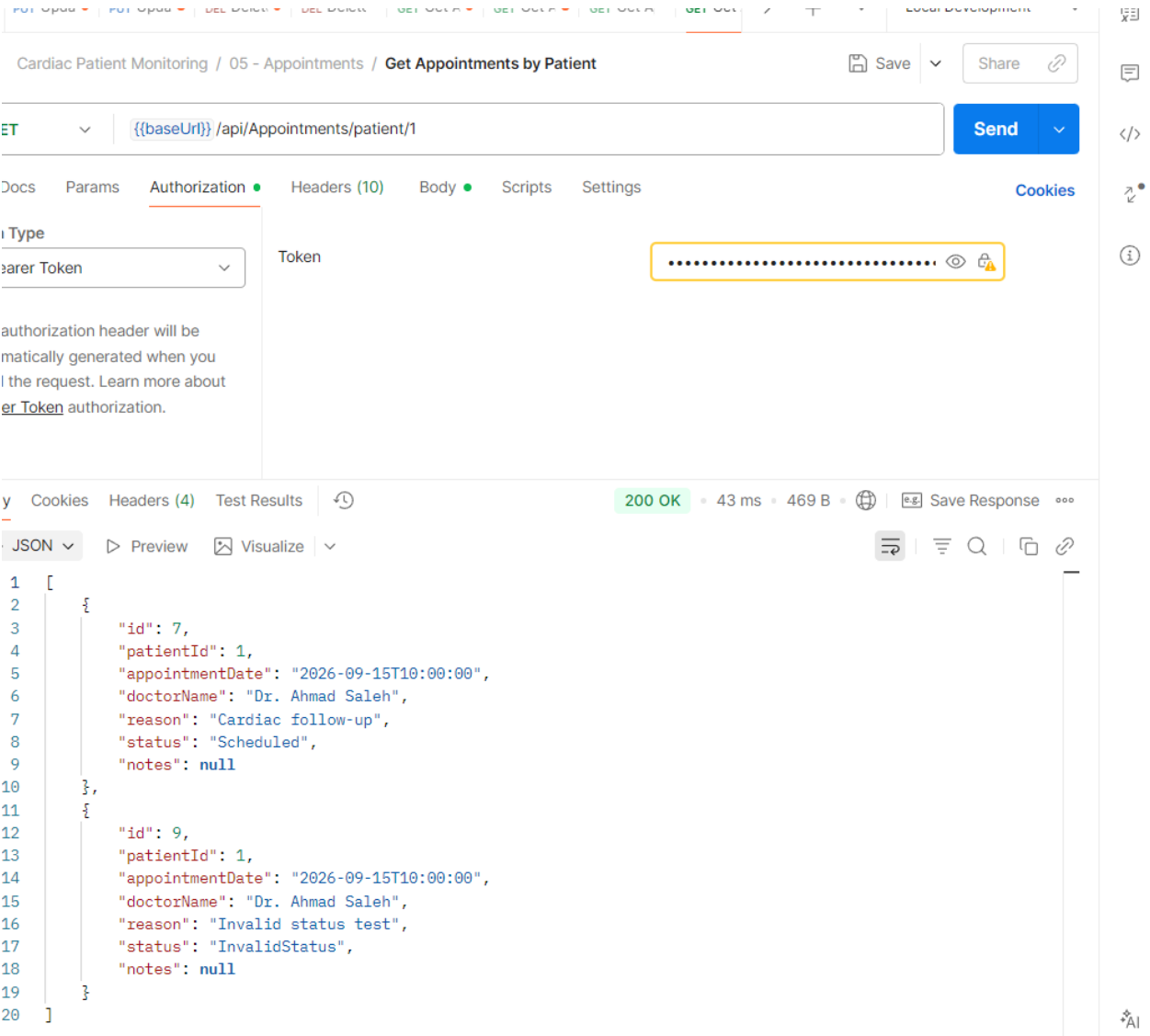
1  [
2    {
3      "id": 7,
4      "patientId": 1,
5      "appointmentDate": "2026-09-15T10:00:00",
6      "doctorName": "Dr. Ahmad Saleh",
7      "reason": "Cardiac follow-up",
8      "status": "Scheduled",
9      "notes": null
10   },
11   {
12     "id": 9,
13     "patientId": 1,
14     "appointmentDate": "2026-09-15T10:00:00",
15     "doctorName": "Dr. Ahmad Saleh",
16     "reason": "Invalid status test",
17     "status": "InvalidStatus",
18     "notes": null
19   }
20 ]

```

TC-53 Filter appointments by Status

Endpoint	GET /api/Appointments?status={status}
Module	Appointments
Rule / Behavior Tested	Filters correctly by Status
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot



TC-54 Filter appointments by DoctorName

Endpoint	GET /api/Appointments?doctorName={name}
Module	Appointments
Rule / Behavior Tested	Filters correctly by DoctorName
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / Get Appointments by Patient

ET Send

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Type
Bearer Token

Token
.....

authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

200 OK • 43 ms • 469 B • Save Response

JSON Preview Visualize

```

1  [
2    {
3      "id": 7,
4      "patientId": 1,
5      "appointmentDate": "2026-09-15T10:00:00",
6      "doctorName": "Dr. Ahmad Saleh",
7      "reason": "Cardiac follow-up",
8      "status": "Scheduled",
9      "notes": null
10   },
11   {
12     "id": 9,
13     "patientId": 1,
14     "appointmentDate": "2026-09-15T10:00:00",
15     "doctorName": "Dr. Ahmad Saleh",
16     "reason": "Invalid status test",
17     "status": "InvalidStatus",
18     "notes": null
19   }
20 ]

```

TC-55 Get appointments for existing patient

Endpoint	GET /api/Appointments/patient/{patientId}
Module	Appointments
Rule / Behavior Tested	Returns appointments for a valid patient
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / Get Appointments by Patient

ET Send

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Type
Bearer Token

Token

authorization header will be automatically generated when you click the request. Learn more about [Bearer Token](#) authorization.

200 OK • 43 ms • 469 B • Save Response

JSON Preview Visualize

```

1  [
2    {
3      "id": 7,
4      "patientId": 1,
5      "appointmentDate": "2026-09-15T10:00:00",
6      "doctorName": "Dr. Ahmad Saleh",
7      "reason": "Cardiac follow-up",
8      "status": "Scheduled",
9      "notes": null
10   },
11   {
12     "id": 9,
13     "patientId": 1,
14     "appointmentDate": "2026-09-15T10:00:00",
15     "doctorName": "Dr. Ahmad Saleh",
16     "reason": "Invalid status test",
17     "status": "InvalidStatus",
18     "notes": null
19   }
20 ]

```

TC-56 Get appointments for non-existing patient

Endpoint	GET /api/Appointments/patient/{patientId}
Module	Appointments
Rule / Behavior Tested	Patient ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

[HTTP](#) Cardiac Patient Monitoring / 05 - Appointments / **Get Appointments by Patient — Patient Not Found**

Save

Share

GET

{{baseUrl}}/api/Appointments/patient/9999

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

Auth Type

Bearer Token

Token

.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body

Cookies

Headers (4)

Test Results

404 Not Found

12 ms

204 B

Save Response

{}

JSON

Preview

Debug with AI

1

2

3

{

"message": "Patient with ID 9999 was not found."

}

TC-57 Get existing appointment

Endpoint	GET /api/Appointments/{id}
Module	Appointments
Rule / Behavior Tested	Returns appointment by valid ID
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / **Get All Appointments** Save Share

GET `{{baseUrl}}/api/Appointments/2` Send

Docs Params **Authorization** Headers (10) Body Scripts Settings Cookies

Auth Type
Bearer Token ▼

Token 👁 🔒

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (4) **Test Results** 🔄 200 OK 13 ms 312 B 🌐 💾 Save Response ⋮

{} JSON ▼ ▶ Preview 🖼 Visualize ▼ 🔍 📄 🔗

```

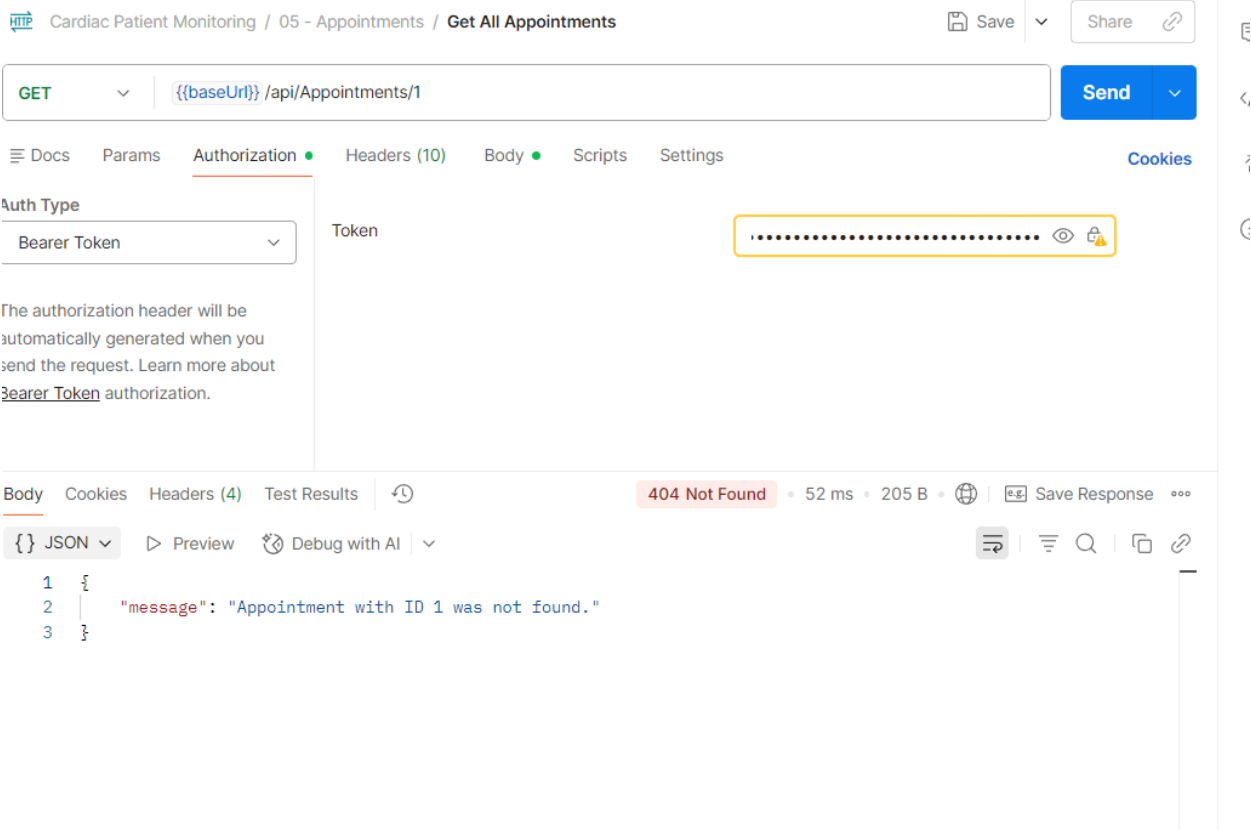
1 {
2   "id": 2,
3   "patientId": 2,
4   "appointmentDate": "2026-09-20T10:00:00",
5   "doctorName": "Dr. Ahmad Saleh",
6   "reason": "Updated cardiac follow-up",
7   "status": "Scheduled",
8   "notes": null
9 }

```

TC-58 Get non-existing appointment

Endpoint	GET /api/Appointments/{id}
Module	Appointments
Rule / Behavior Tested	Appointment ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot



TC-59 Create valid appointment

Endpoint	POST /api/Appointments
Module	Appointments
Rule / Behavior Tested	Valid appointment data is accepted
Expected Result	201 Created
Actual Result	201 Created — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / Create Appointment - Valid

POST {{baseUrl}}/api/Appointments **Send**

Auth Type: Bearer Token

Token:

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers (5) Test Results 201 Created • 58 ms • 364 B • Save Response

{ } JSON Preview Visualize

```

1 {
2   "id": 10,
3   "patientId": 1,
4   "appointmentDate": "2026-09-15T10:00:00Z",
5   "doctorName": "Dr. Ahmad Saleh",
6   "reason": "Cardiac follow-up",
7   "status": "Scheduled",
8   "notes": null
9 }
```

TC-60 Create appointment for invalid patient

Endpoint	POST /api/Appointments
Module	Appointments
Rule / Behavior Tested	Referenced patient must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot

The screenshot shows a REST client interface for a test titled "Create Appointment - Invalid Patient ID". The request is a POST to the endpoint `{{baseUrl}}/api/Appointments`. The Authorization tab is active, showing "Bearer Token" as the auth type. A token is entered in the field, represented by dots and a masked icon. Below the token field, a note states: "The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization."

The response section shows a "404 Not Found" status with a response time of 10 ms and a body size of 205 B. The response body is displayed in JSON format:

```
1 {  
2   "message": "Patient with ID 99999 was not found."  
3 }
```

TC-61 Create appointment with missing required field

Endpoint	POST /api/Appointments
Module	Appointments
Rule / Behavior Tested	Required fields must be present
Expected Result	400 Bad Request
Actual Result	400 Bad Request — PASS

Evidence / Screenshot

Cardiac Patient Monitoring / 05 - Appointments / **Create Appointment - Missing DoctorName** Save Share

POST ▼ `{{baseUrl}}/api/Appointments` Send

Docs Params Authorization Headers (10) Body Scripts Settings Cookies

Auth Type
Be... ▼

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token

Body Cookies Headers (4) Test Results 400 Bad Request 6 ms 446 B Save Response

{} JSON Preview Debug with AI

```

1  {
2    "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
3    "title": "One or more validation errors occurred.",
4    "status": 400,
5    "errors": {
6      "DoctorName": [
7        "The DoctorName field is required.",
8        "Doctor name is required."
9      ]
10   },
11   "traceId": "00-2c49806c75cd7c9a6c59ce81bc9bf695-3c739820575048ee-00"
12 }

```

TC-62 Update existing appointment

Endpoint	PUT /api/Appointments/{id}
Module	Appointments
Rule / Behavior Tested	Valid update data is accepted
Expected Result	200 OK
Actual Result	200 OK — PASS

Evidence / Screenshot

[HTTP](#) Cardiac Patient Monitoring / 05 - Appointments / **Update Appointment - Not Found**

Save

Share

PUT

{{baseUrl}}/api/Appointments/2

Send

Docs

Params

Authorization

Headers (10)

Body

Scripts

Settings

Cookies

Auth Type

Bearer Token

Token

.....

he authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body

Cookies

Headers (4)

Test Results

200 OK

74 ms

302 B

Save Response

{ } JSON

Preview

Visualize

1 {

2 "Id": 2,

3 "patientId": 2,

4 "appointmentDate": "2026-09-20T10:00:00Z",

5 "doctorName": "Dr. Ahmad Saleh",

6 "reason": "Not found test",

7 "status": "Scheduled",

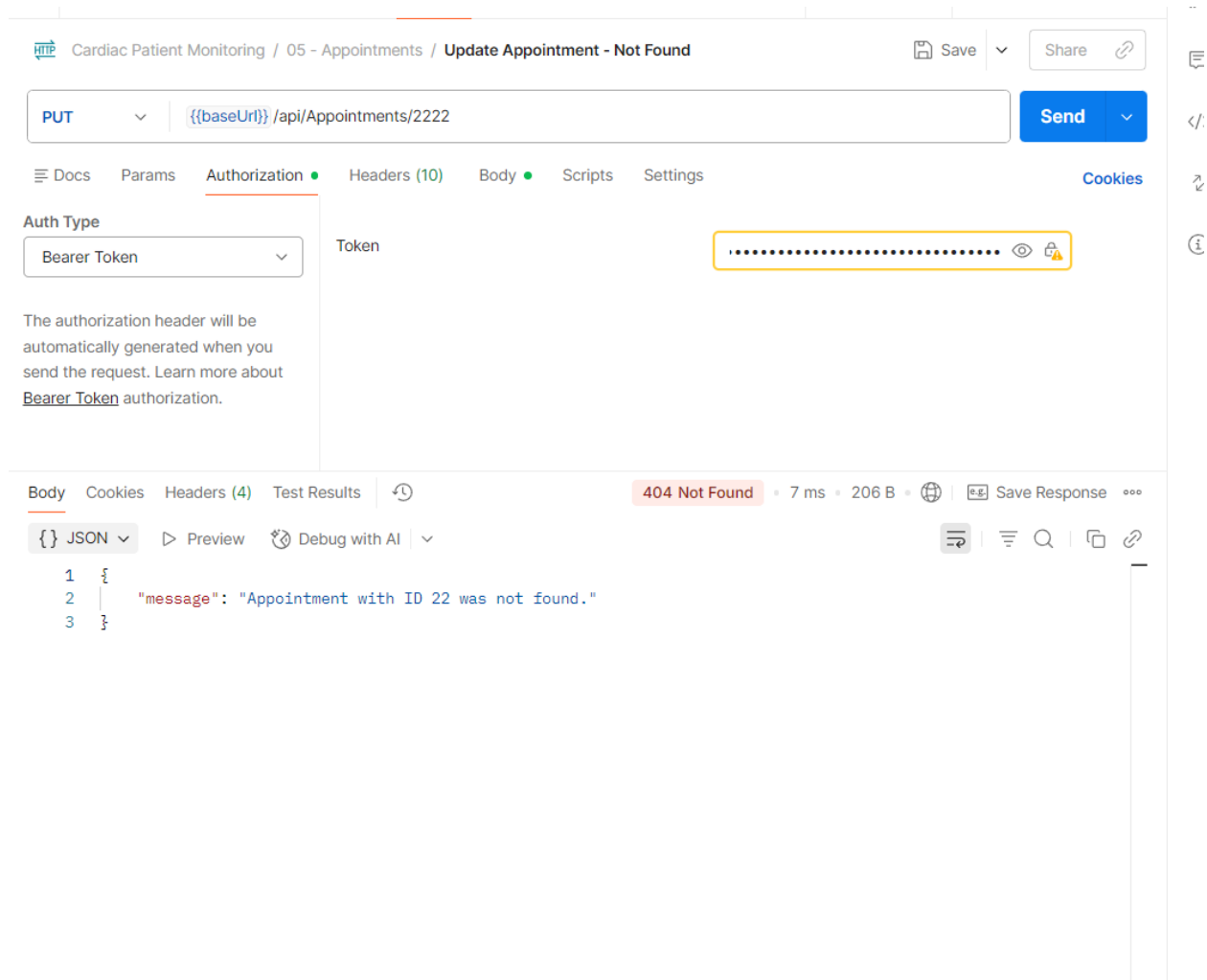
8 "notes": null

9 }

TC-63 Update non-existing appointment

Endpoint	PUT /api/Appointments/{id}
Module	Appointments
Rule / Behavior Tested	Appointment ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot



Cardiac Patient Monitoring / 05 - Appointments / Update Appointment - Not Found

PUT `{{baseUrl}}/api/Appointments/2222` **Send**

Auth Type: Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token:

Body Cookies Headers (4) Test Results **404 Not Found** 7 ms 206 B Save Response

```
{
  "message": "Appointment with ID 22 was not found."
}
```

TC-64 Delete existing appointment

Endpoint	DELETE /api/Appointments/{id}
Module	Appointments
Rule / Behavior Tested	Valid ID is deleted
Expected Result	204 No Content
Actual Result	204 No Content — PASS

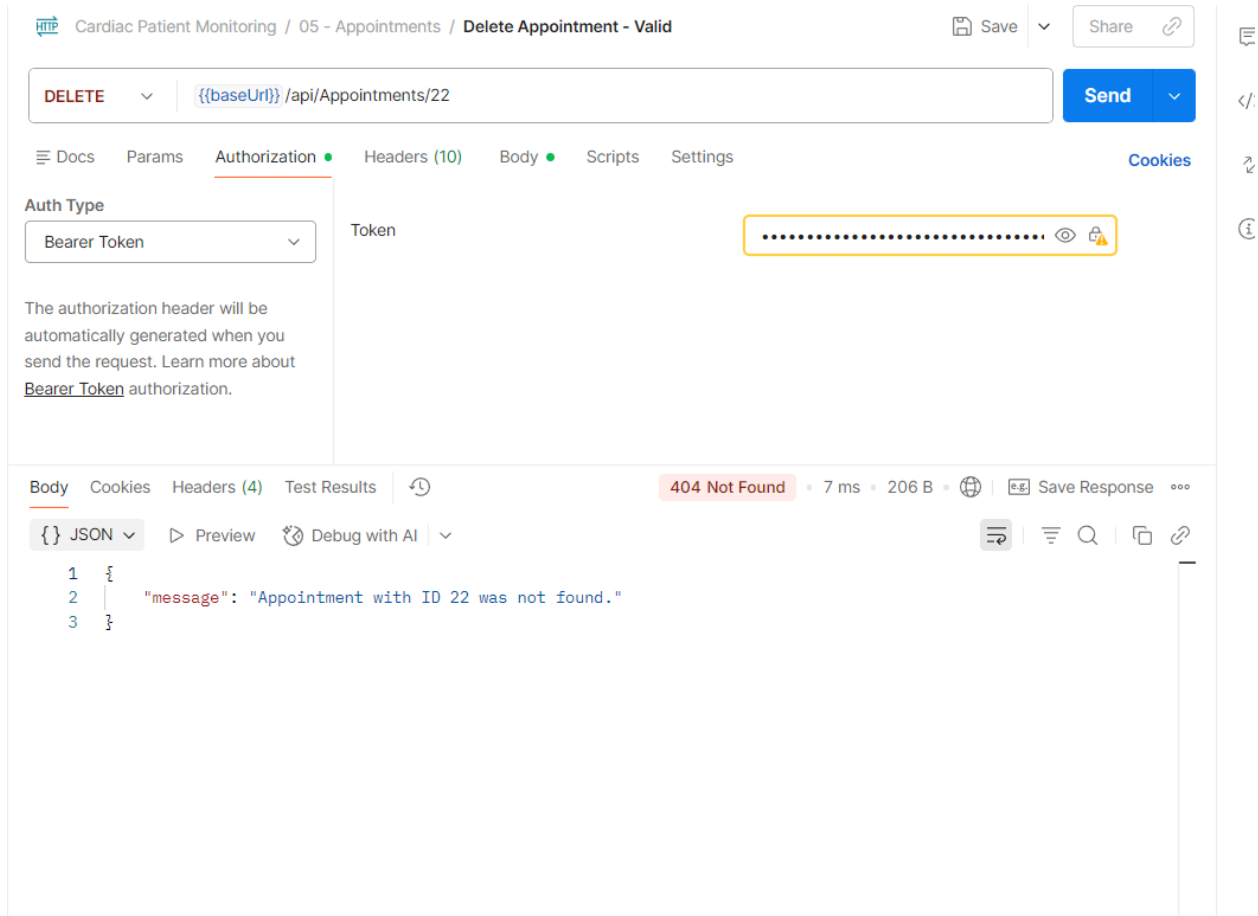
Evidence / Screenshot

The screenshot shows a REST client interface for a test titled "Delete Appointment - Valid". The request is a DELETE to the endpoint `{{baseUrl}}/api/Appointments/2`. The authorization is set to "Bearer Token". The response is "204 No Content" with a status of 204, a response time of 40 ms, and a body size of 81 B. The interface includes tabs for Docs, Params, Authorization, Headers (10), Body, Scripts, and Settings. The Authorization tab is active, showing the Bearer Token field. The response tab is also active, showing the 204 No Content status.

TC-65 Delete non-existing appointment

Endpoint	DELETE /api/Appointments/{id}
Module	Appointments
Rule / Behavior Tested	Appointment ID must exist
Expected Result	404 Not Found
Actual Result	404 Not Found — PASS

Evidence / Screenshot



Cardiac Patient Monitoring / 05 - Appointments / Delete Appointment - Valid

DELETE Send

Auth Type: Bearer Token

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Token:

Body: 404 Not Found • 7 ms • 206 B

```
{
  "message": "Appointment with ID 22 was not found."
}
```

6. Sample Structured Error Responses

Two issues were identified during testing where the API initially returned 200 OK for invalid data. Both were fixed and retested; sample structured responses after the fix are shown below.

6.1 Vital Signs — Invalid Heart Rate

Creating a vital sign record with HeartRate = 500 initially returned 200 OK. After adding range validation, the API correctly returns:

```
{
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": {
    "HeartRate": [
      "Heart rate is outside the valid physiological range."
    ]
  }
}
```

6.2 Medications — Missing StartDate

Creating a medication without a StartDate initially defaulted the value to 0001-01-01T00:00:00 and returned 200 OK. After tightening DTO validation, the API correctly returns:

```
{
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": {
    "StartDate": [
      "StartDate is required."
    ]
  }
}
```

7. Test Summary

Module	Test Cases	Passed
Register	5	5
Login	4	4
JWT	3	3
Current User	2	2
Authorization	2	2
Patients	9	9
Vital Signs	12	12
Medications	13	13
Appointments	15	15

Total Test Cases: 65

Passed: 65

Failed: 0

8. Conclusion

All 65 test cases executed against the CardiacPatientMonitoringApi returned the expected HTTP status codes, confirming that authentication, authorization, and CRUD/validation logic behave correctly across the Patients, Vital Signs, Medications, and Appointments modules.

Two validation gaps were identified and fixed during testing: an out-of-range heart rate that was initially accepted (200 OK) and a missing medication StartDate that was silently defaulted instead of rejected. Both were corrected and retested successfully, confirming the API now returns 400 Bad Request for these invalid inputs.

This validates the current testing foundation and confirms the API is ready for the next stage of automated testing using xUnit, WebApplicationFactory, and Moq, targeting the same success and failure scenarios already verified manually.

— end of Validation Testing Report —